

Quantum Cryptography

RSA Keys

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Team Fun(Key)

RSA

- Secret prime numbers p, q
- $N = pq$
- Secret totient $\theta(n) = \text{lcm}(p-1, q-1)$
- E is $1 < e < \theta(n)$
- $D * e = 1 \bmod \theta(n)$

Physical RSA

- Common implementation of 2-Factor Authentication
- Common use is within (Password|Token) to login to stations
- Problem: Quantum-Safe Tokens



Physical RSA

- Assume a 12 character length password (Upper, lower, special, etc)
- NIST recommends at least 3072-bit keys for quantum resistant keys (We will assume 4096 bits for simplicity 2^{12})
- $12 \times 8 = 96$ bits reserved, 4000 bits left for key, key range $[0, 2^{4000}-1]$

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11424435476401506867567044883383791814261499396141697332137629902095226710
31123194426911850674917242872214865561662124663332057533500850377750025412
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84243782302841753466779571476555456024747173876881477665773192923705560212
82569709497965687274449565001974078442788749994488443682110924676873252627
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```

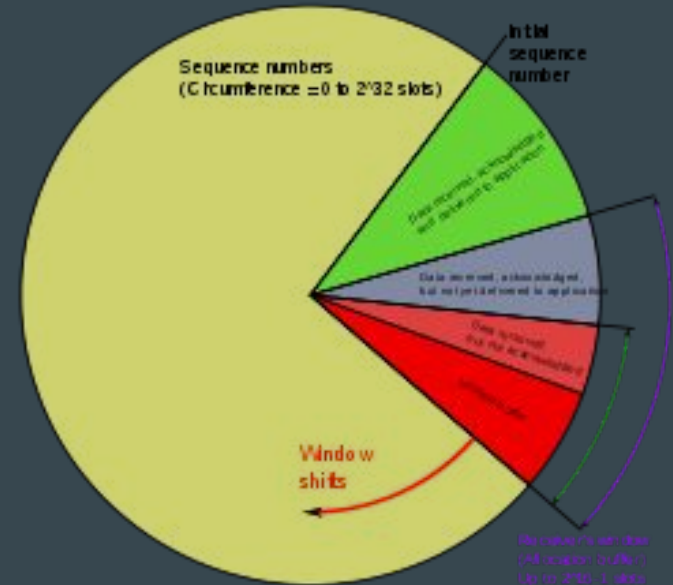
Issues

- Birthday Paradox
- Hardware limitations
- Establishing a large key



Packet Fragmentation and RSA

- IPv4 (Supports fragmentation)
- IPv6 (Does not require fragmentation header)
- Noisy Networks
- Forced packet fragmentation
- TCP Attacks



RSA SSL/TLS

- Client and Server need to communicate in order to exchange encryption capabilities.
- Client first authenticates server with the server certificate.
- This certificate also includes the server's RSA public key, but asymmetric encryption isn't ideal for web communications.
- Client will use the public key to share the methods and secret values necessary for generating a mutual symmetric key for both ends of the connection.
- If the confidentiality of the RSA key exchange is broken, any eavesdropper can generate the symmetric key and read any transmitted information.

Effects of stronger RSA in SSL

- Keeping the data confidential during a SSL/TLS handshake can be as simple as increasing the RSA key size.
- Currently the recommended key size is around 2048 bits. Most major web services utilize this key size.
- However, increasing the key size could result in slower connections, and increased processing on the server side.
- Might not make a huge impact in single cases, but what happens if hundreds of thousands of clients make a request every second?

Quantum Computers vs RSA

- Breaking RSA key size of 2048 bits requires 4000 qubits estimatedly.
- By contrast, current quantum computers are capable of pushing only upto 72 qubits.
- Perfect qubits with 1:1 physical ratio that could factor RSA 2048 could be pushed earliest by 2023.

Quantum Safe RSA

- For now implementing key sizes of 3072 bits or larger gives us an ample amount of time
- RSA keys of size 4096 bits are essentially “Quantum Safe” for now

RSA VPN

- Most VPNs use RSA 2048 bit keys in the Handshake.
- Will be rendered unsalvageable with development of Quantum Computers.
- Possible solution is to use 4096 bit keys instead.
- Increases encryption strength by a factor of 2^{18} .

Issues

- The SSL handshake at the start of each connection will be slower.
- There's an increase in CPU usage during handshakes.
- Compatibility concerns. Neither Cisco IOS nor Amazon Cloudfront support 4096 bit keys.

			sign	verify	sign/s	verify/s
rsa	512	bits	0.000210s	0.000014s	4772.1	69667.5
rsa	1024	bits	0.000727s	0.000035s	1375.3	28508.9
rsa	2048	bits	0.003778s	0.000092s	264.7	10899.5
rsa	4096	bits	0.022637s	0.000305s	44.2	3275.4

Post Quantum VPN

- VPNs based on Post Quantum Cryptography Algorithms.
- Implemented using algorithms in NIST's Round 2 submissions of Post Quantum Cryptography project
- Frodo, SIKE and Picnic implemented in Microsoft's PQ VPN project.
- Still require considerable amount of analysis to judge performance and practicality.
- Are they an improvement over simply using larger RSA keys?

Sources

- <https://cryptome.org/2016/01/CNSA-Suite-and-Quantum-Computing-FAQ.pdf>
- <https://certsimple.com/blog/measuring-vpn-rsa-keys>
- <https://www.microsoft.com/en-us/research/project/post-quantum-crypto-vpn/>